TULSIRAMJI GAIKWAD-PATIL College of Engineering and Technology
Wardha Road, Nagpur - 441108
Accredited with NAAC A+ Grade
Approved by AICTE, New Delhi, Govt. of Maharashtra

(An Autonomous Institute Affiliated to RTM Nagpur University with NAAC A+ Grade)
Department of Information Technology
Session 2023-24(Odd Semester)

VISION	MISSION
To emerge as a learning hub and center of excellence in the domain of Information Technology	[M1] To impart quality technical education through effective teaching learning process. [M2] To provide a platform to address societal issues as well as challenges faced by IT industries. [M3] To foster a culture of research and impart innovative and entrepreneurial skills in the field of IT. [M4] To ensure overall development of students and staff by inculcating knowledge and professional ethics as a part of lifelong learning.

EndSemester
Semester: III
Programme: B. Tech Information Technology
Course: Data Structure

Time: 3 Hours **Max Marks:** 60 Marks

Instructions to candidates:

1. All questions are compulsory in Group A
2. Solve any two sub-question out of three in Group B
3. Assume suitable data wherever necessary
4. All question carries marks as indicated

BIT32301	Course Outcome
CO1	To gain knowledge about basic concepts of data structures.
CO2	To acquire knowledge of stacks and queues with their applications.
CO3	To aware about the concepts of trees with their applications.
CO4	To learn and design the algorithm for graphs with their applications.
CO5	To implement basic operations on linked list.

Group A				
Que. 1	Answer the following Question	CO	BTL	Marks
a.	Define Data Structure.	CO1	2	2
b.	What is a Null Pointer?	CO2	2	2
c.	Define Circular Queue.	CO3	2	2
d.	Define BFS Traversal.	CO4	3	2
e.	Define Complete Binary Tree.	CO5	3	2
Group B				
Que. 2	Answer the following Question (Solve any Two)	CO	BTL	Marks

a.	Explain Linear Search and Binary Search with examples	CO1	3	5
b.	Explain Quick Sort algorithm with suitable example.	CO1	3	5
c.	Compare Bubble Sort, Selection Sort and Insertion Sort.	CO1	4	5
Que. 3	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain Types of Linked List with neat diagram.	CO2	4	5
b.	Describe insertion and deletion operations in Linked List.	CO2	3	5
c.	Compare Singly, Doubly and Circular Linked List.	CO2	2	5
Que. 4	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain Stack representation using Array.	CO3	2	5
b.	Explain Queue operations: Insert and Delete.	CO3	4	5
c.	Explain Expression Evaluation using Stack with example.	C03	3	5
Que 5	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain Graph representation using Adjacency List and Matrix.	CO4	2	5
b.	Explain BFS and DFS traversal techniques with examples.	CO4	3	5
c.	Explain Kruskal’s Algorithm for Minimum Spanning Tree.	CO4	3	5
Que 6.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain Binary Tree Traversals with example.	CO5	2	5
b.	Describe BST insertion and deletion operations.	CO5	3	5
c.	Explain AVL Tree rotations with suitable example.	CO5	3	5

Bloom’s Taxonomy Level (BTL)- Remember , Understand , Apply, Analyze , Evaluate , Create